

# IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM Granite and LangFlow”*

# Project Tile – TravelBot: AI-Powered Smart Travel Planner

## Domain of Project – Travel & Tourism

| Student Name    | Recent Passport Photo                                                                | Email ID                 | Mobile number [WhatsApp] |
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# Problem Statement -

## Problem Statement: Travel Planner Agent

### The Challenge

Planning a trip can be time-consuming and overwhelming due to scattered information about destinations, accommodations, transportation, weather, and travel costs. Travelers often struggle to create personalized itineraries that match their budget, interests, and schedule. Without a single intelligent assistant, organizing a smooth and enjoyable travel experience becomes difficult.

### The Objective

Build an AI-powered Travel Planner Agent that provides personalized travel recommendations and trip planning for individuals, families, and groups. The solution should:

- Personalized Trip Planning – Generate customized travel itineraries based on destination, budget, travel duration, interests, and travel style.
- Destination Knowledge Retrieval – Retrieve and summarize information about tourist attractions, local culture, weather, transportation, and travel tips from reliable sources.
- Budget Planning & Cost Estimation – Estimate travel expenses including accommodation, transportation, food, sightseeing, and miscellaneous costs.
- Hotel & Transportation Recommendations – Suggest suitable hotels and transportation options based on user preferences and budget.
  - Multi-Agent System
    - Destination Information Agent – Retrieves destination details and tourist attractions.
    - Itinerary Planning Agent – Generates personalized day-wise travel plans.
    - Budget Planning Agent – Calculates estimated travel expenses.
    - Travel Assistance Agent – Provides travel tips, weather guidance, packing suggestions, and local recommendations.
- Travel Dashboard – Display itinerary summary, budget breakdown, recommended attractions, hotel suggestions, and trip overview through an interactive dashboard.

## Proposed Solution -

### Proposed Solution: AI-Powered Travel Planner Agent

The proposed solution is an AI-powered Travel Planner Agent built using IBM watsonx.ai, IBM Granite Models, and LangFlow to provide personalized travel planning. The system generates customized itineraries, estimates travel budgets, and recommends tourist attractions, hotels, transportation, and travel tips based on user preferences, destination, budget, and trip duration.

The solution includes four intelligent agents:

- Destination Knowledge Agent – Retrieves destination details, weather, and tourist attractions.
- Itinerary Planning Agent – Generates personalized day-wise travel plans.
- Budget Planning Agent – Estimates travel expenses and provides a budget breakdown.
- Travel Assistance Agent – Recommends hotels, transportation, restaurants, and travel tips.

A user-friendly dashboard allows users to interact with the AI assistant, generate travel plans, and view budget summaries. IBM watsonx.ai and Granite Models power the AI responses, while LangFlow manages the overall agent workflow, providing an intelligent and seamless travel planning experience.

# Technology Used

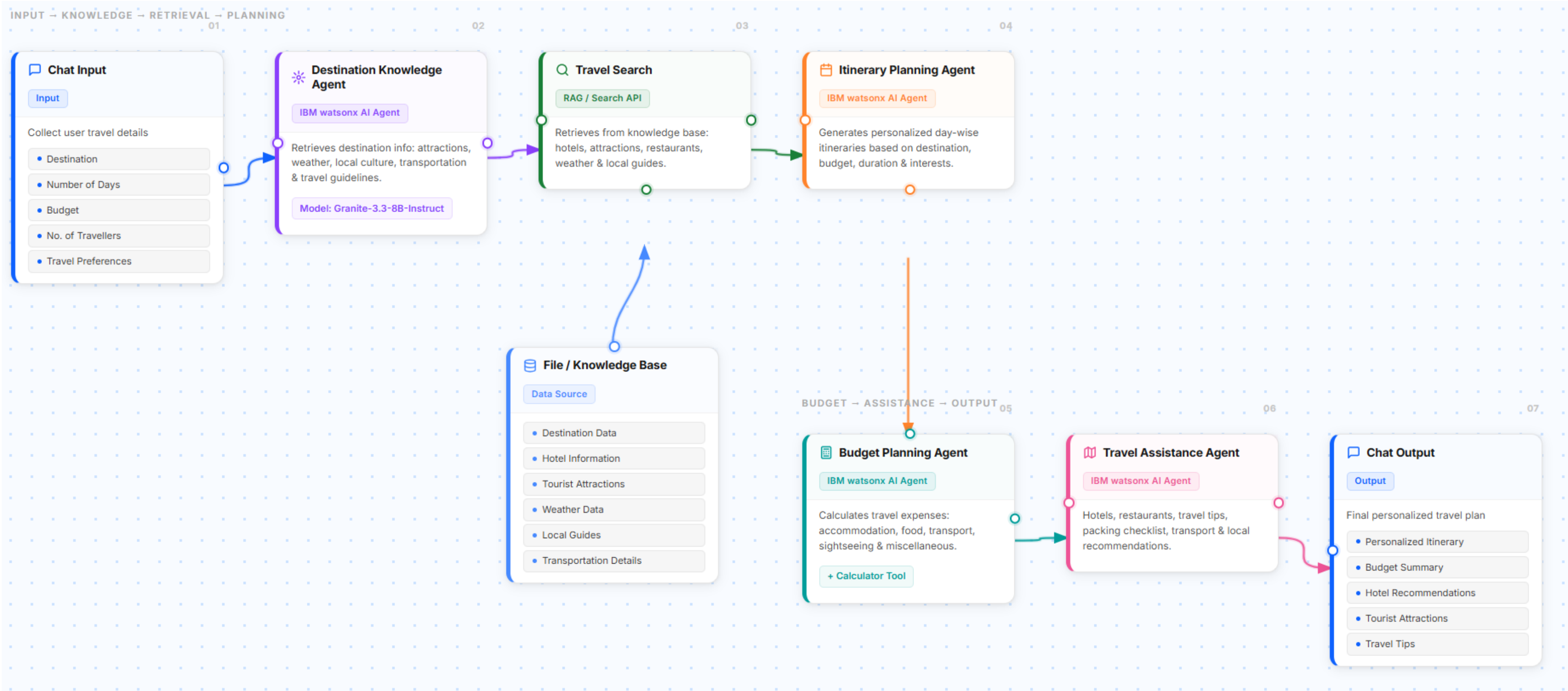
- LangFlow Platform – For designing and orchestrating the multi-agent workflow visually.
- IBM Granite Model (e.g., ibm-granite-3-3-8b-instruct) – For natural language understanding, itinerary generation, reasoning, and personalized travel recommendations.
- IBM Cloud Lite – Provides a secure, scalable cloud platform for developing, deploying, and managing the AI-powered Travel Planner Agent.
- IBM watsonx.ai – For AI model deployment, prompt engineering, model inference, and AI governance.
- Retrieval-Augmented Generation (RAG) – Retrieves destination-specific travel information before generating accurate and context-aware travel recommendations.
- Vector Database (FAISS/ChromaDB) – Stores and retrieves travel knowledge such as tourist attractions, hotels, restaurants, and travel guides for RAG.
- Python Flask – Backend framework used to build the Travel Planner application and APIs.
- Bootstrap 5, HTML, CSS, JavaScript – Used to create a responsive and interactive user interface.



## Langflow component Used -

- 1) Chat Input – Interface where users enter travel details such as destination, budget, travel duration, number of travellers, and travel preferences.
- 2) IBM watsonx AI Agent – Processes user input using IBM Granite models to generate personalized travel recommendations, itineraries, and travel guidance.
- 3) IBM Granite Model – Granite-3.3-8B-Instruct (or the Granite model you actually use) – Performs natural language understanding, itinerary generation, travel reasoning, and personalized trip planning.
- 4) Chat Output – Displays AI-generated travel itineraries, destination recommendations, budget estimates, hotel suggestions, and travel tips.
- 5) File Component – Stores and retrieves travel knowledge such as destination guides, tourist attractions, hotel information, transportation details, and travel documents for AI processing.
- 6) Prompt Component – Formats user queries and travel preferences into structured prompts before sending them to the IBM Granite model for accurate and context-aware responses.

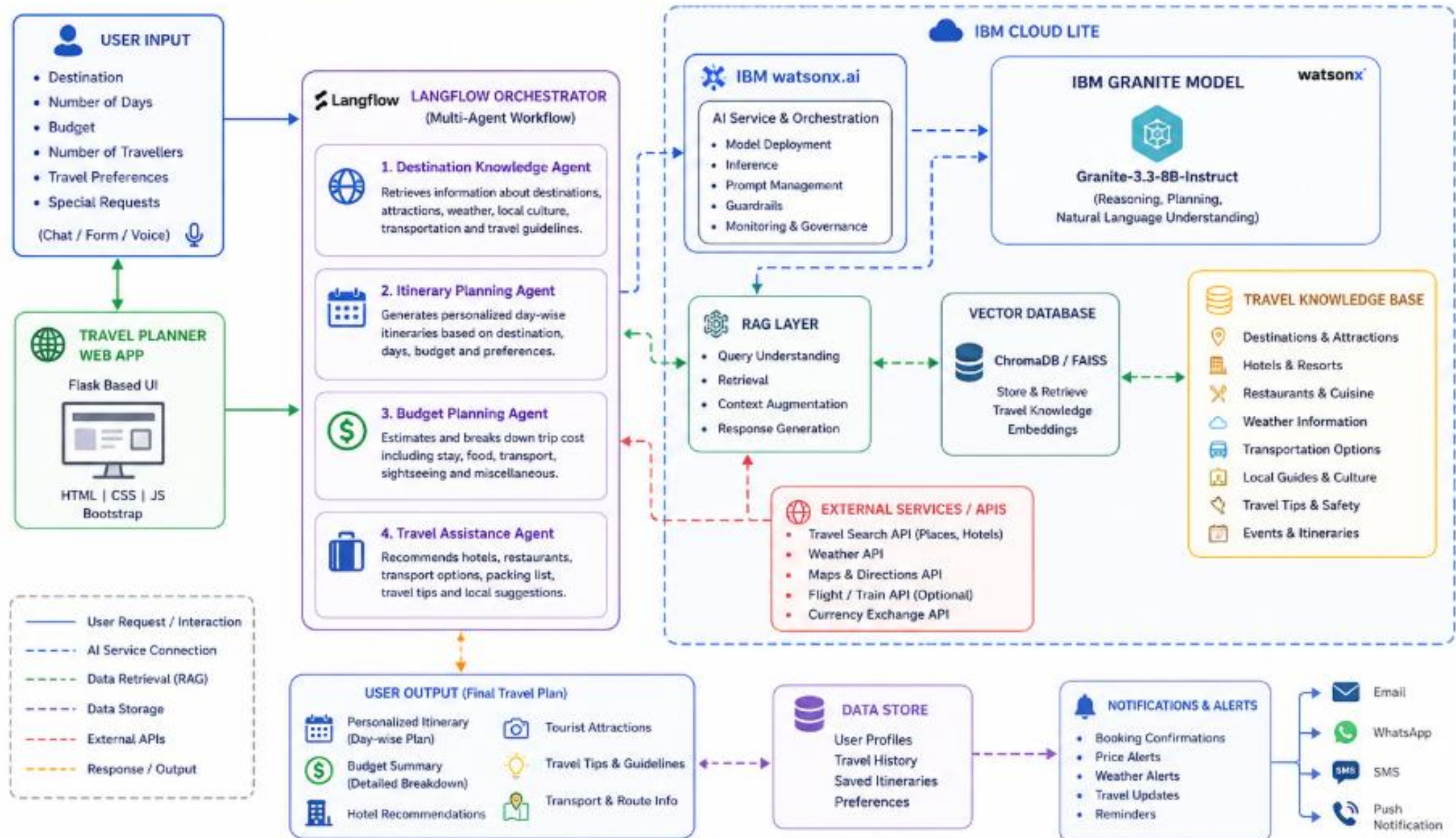
# Langflow workflow - Langflow-based Multi-Agent Workflow of the Smart Travel Planner Agent





# Smart Travel Planner Agent – Architecture Blueprint

AI-Powered Personalized Travel Planning using IBM watsonx.ai, Granite Models & Langflow

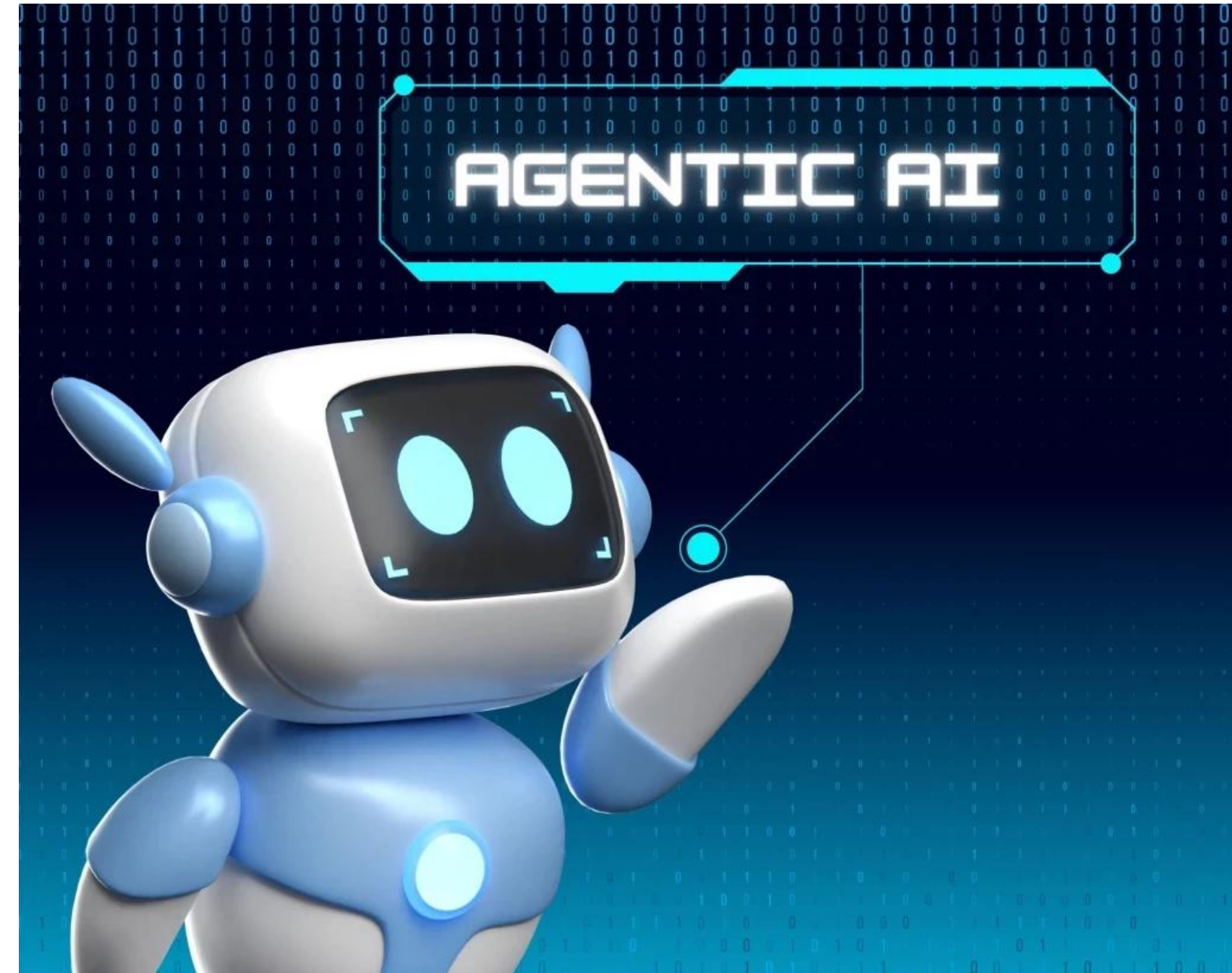


This architecture leverages IBM Cloud, IBM watsonx.ai, Granite Models and Langflow to deliver an intelligent, reliable and personalized travel planning experience.



# Role of Agentic AI in the solution

The Agentic AI system enables the Travel Planner Agent to autonomously understand user requirements, retrieve destination-specific information, and coordinate multiple specialized agents to generate personalized travel plans. Each agent performs a dedicated task, such as destination knowledge retrieval, itinerary generation, budget estimation, and travel assistance, while collaborating to deliver accurate, context-aware, and user-centric recommendations. This multi-agent approach improves decision-making, automation, and overall travel planning efficiency.



## Novelty and Uniqueness

- AI-powered personalized travel planning using IBM watsonx.ai and IBM Granite Models.
- Multi-agent architecture for destination search, itinerary generation, budget planning, and travel assistance.
- Dynamic, destination-specific itineraries based on user preferences, budget, and travel duration.
- Intelligent budget estimation with a detailed breakdown of accommodation, transport, food, and sightseeing costs.
- Context-aware recommendations for hotels, tourist attractions, restaurants, and local travel tips.
- Scalable cloud-based solution built on IBM Cloud Lite for reliable and efficient AI services.

## Git Hub Link

<https://github.com/mahathi137/TravelPlannerBot>

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# Future Scope

## 1. Real-Time Booking Integration

Integrate flight, hotel, and transportation booking APIs to enable users to search, compare, and book travel services directly from the Travel Planner Agent.

## 2. AI-Based Personalized Recommendations

Use user preferences, travel history, and real-time data to provide personalized destination suggestions, optimized itineraries, and smart budget recommendations for every trip.

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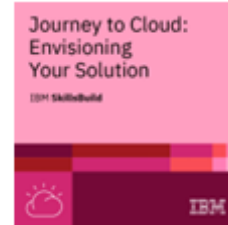
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# Thank You!

Thank you for your time and interest.